

Kernel Density Estimation

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This document summarises the main results on kernel density estimation that are relevant to our work. For more details, see Chacón & Duong (2018) and the references therein.

Given a set of observations $\mathbf{y}_i \in \mathbb{R}^m$ for $i \in \{1, \dots, n\}$, a kernel density estimate is defined as

$$\hat{f}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2}(\mathbf{y} - \mathbf{y}_i)), \quad (1)$$

where K is a square-integrable spherically-symmetric function, bounded below by 0, with a finite second-order moment and unit integral, and $\mathbf{H}$ is a symmetric $m \times m$ positive-definite matrix.

A standard way to measure the performance of a density estimator is via the mean integrated squared error (MISE) defined as

$$\text{MISE}(\hat{f}(\cdot, \mathbf{H})) = \mathbb{E} \left[\int \left(\hat{f}(\mathbf{u}) - f(\mathbf{u}) \right)^2 d\mathbf{u} \right]$$

That is, MISE is the expected L_2 distance between the estimator $\hat{f}$ and the true density f .

We say that $\hat{f}$ is a *consistent* estimator of f if $\text{MISE}(\hat{f}) \rightarrow 0$ as $n \rightarrow \infty$.

We will need some conditions on the density function, the kernel and the bandwidth matrix to ensure consistency of the kernel density estimator. These are given below.

- A1** The density function f is square integrable and twice differentiable, with all of its second-order partial derivatives bounded, continuous and square integrable.
- A2** The kernel K is a square-integrable, spherically symmetric function on $\mathbb{R}^m$ that satisfies $\int K(\mathbf{u}) d\mathbf{u} = 1$, $\int \mathbf{u} K(\mathbf{u}) d\mathbf{u} = \mathbf{0}$ and $\int \mathbf{u} \mathbf{u}^T K(\mathbf{u}) d\mathbf{u} = \mu_2(K) \mathbf{I}_m$ for some $\mu_2(K) > 0$.
- A3** The bandwidth matrix $\mathbf{H}$ is a sequence of symmetric positive-definite matrices such that $\mathbf{H} \rightarrow \mathbf{O}$ and $n|\mathbf{H}|^{1/2} \rightarrow \infty$ as $n \rightarrow \infty$.

Theorem 1 (Consistency) *Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be a sequence of independent random variables from a distribution on $\mathbb{R}^m$ that satisfies A1. Then under conditions A2 and A3,*

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int \left(\hat{f}(\mathbf{u}) - f(\mathbf{u}) \right)^2 d\mathbf{u} \right] = 0.$$

An asymptotic approximation of the MISE is given by

$$\text{AMISE}(\hat{f}(\cdot, \mathbf{H})) = n^{-1} |\mathbf{H}|^{-1/2} R(K) + \frac{1}{4} \mu_2(K)^2 \int \{\text{tr}(\mathbf{H} D^2 f(\mathbf{u}))\}^2 d\mathbf{u}, \quad (2)$$

where $R(K) = \int K(\mathbf{u})^2 d\mathbf{u}$ and $D^2 f$ is the Hessian of f .

If $\mathbf{H} = h_n \mathbf{I}_m$ for some $h_n > 0$, then we can write

$$\text{AMISE}(\hat{f}(\cdot, h_n \mathbf{I}_m)) = n^{-1} h_n^{-m/2} R(K) + \frac{1}{4} \mu_2(K)^2 h_n^2 \int \{\text{tr}(D^2 f(\mathbf{u}))\}^2 d\mathbf{u}. \quad (3)$$

The minimal MISE rate is achieved when h_n is of order $n^{-2/(m+4)}$, and the corresponding MISE is of order $n^{-4/(m+4)}$.

However, we are not really interested in optimizing the MISE, as we care more about the tails of the distribution where f is relatively small.

References

Chacón, J. E. & Duong, T. (2018), *Multivariate kernel smoothing and its applications*, CRC Press, Boca Raton, Florida.